a Direct training

brain response only, no scene features

fMRI response (370 ROIs, participant *s*)

↓

participant- and ROI-specific linear layers → ROI tokens

↓

shared encoder → *H*

↓

34 emotion queries read *H*: *g*(*H*)

↓

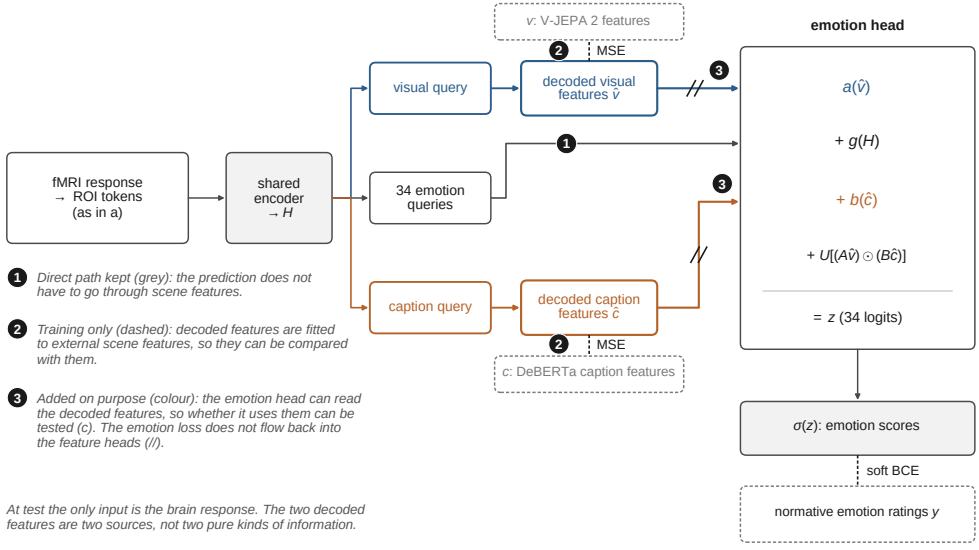
σ(*z*): 34 emotion scores

soft BCE

normative emotion ratings *y* (34 proportions, 0 to 1)

b Decoded-feature readout: paths added to test a question

Does the emotion prediction use brain signals that correspond to scene features?

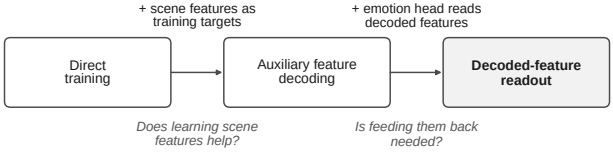


1. *Direct path kept (grey): the prediction does not have to go through scene features.*
2. *Training only (dashed): decoded features are fitted to external scene features, so they can be compared with them.*
3. *Added on purpose (colour): the emotion head can read the decoded features, so whether it uses them can be tested (c). The emotion loss does not flow back into the feature heads (//).*

At test the only input is the brain response. The two decoded features are two sources, not two pure kinds of information.

c Checking the added paths

Does each added step help the prediction? (performance against normative emotion ratings)



Other comparisons (same encoder, one change each)

Interaction vs additive head (within readout)	*Do the two decoded features combine?*
Response-based distillation vs direct training	*Are the predictor's scores alone enough?*
Response-based distillation vs label smoothing	*Teacher content, or only softer targets?*
Auxiliary feature decoding vs response-based distillation	*Scene features or scores as targets?*
Readout + response-based distillation vs readout	*Does the teacher add anything on top?*

Response-based distillation mixes the visual-caption predictor's out-of-fold scores into the target only during training; it is not part of the readout condition. Gains are claimed only where an added step beats the simpler model.

Does the model use the added paths? (swap test on test videos)

Video A and its matched video B (nearest in the 73 semantic labels, same test fold). ROI tokens stay at A; only $\hat{v}$ and/or $\hat{c}$ are taken from B.

$\hat{c}$ from \ $\hat{v}$ from	A	B
A	$z(A,A)$ original	$z(B,A)$ visual swapped
B	$z(A,B)$ caption swapped	$z(B,B)$ both swapped

If these values are swapped, does the emotion prediction move toward B's ratings, beyond a random change of the same size?

$I = z(B,B) - z(B,A) - z(A,B) + z(A,A)$ (logits; $I = 0$ for the additive head)

A large effect means this model relies on the path; it does not measure how much of the brain's emotion information the path carries.

Arrows show the model's computation; they do not assume a processing order in the brain.